Height Growth Equations for CONUS: Loblolly Pine

Greg Johnson, David Marshall, Aaron Weiskittel

2026-04-01

Data

We took the FIA height growth data subset for loblolly pine, filtered out stray (likely keypunch errors) states, FIA plot locations with missing EPA L3 Codes, and where the diameter at the end of the 5-year or greater measurement interval was less than the starting diameter. This left 383322 growth observations.

The observations are distributed among L3 Codes and states as follows:

NA_L3NAME	NA_L3CODE	n
Southeastern Plains	8.3.5	134074
South Central Plains	8.3.7	80476
Piedmont	8.3.4	78410
Middle Atlantic Coastal Plain	8.5.1	37364
Southern Coastal Plain	8.5.3	12955
Mississippi Valley Loess Plains	8.3.6	9895
Southwestern Appalachians	8.4.9	7401
Ridge and Valley	8.4.1	7385
Ouachita Mountains	8.4.8	6597
Arkansas Valley	8.4.7	1899
Interior Plateau	8.3.3	1892
Blue Ridge	8.4.4	1131
Mississippi Alluvial Plain	8.5.2	1051
East Central Texas Plains	8.3.8	958
Western Gulf Coastal Plain	9.5.1	803
Boston Mountains	8.4.6	374
Western Allegheny Plateau	8.4.3	188
Interior River Valleys and Hills	8.3.2	141
Texas Blackland Prairies	9.4.7	103

NA_L3NAME	NA_L3CODE	n
Northern Piedmont	8.3.1	100
Ozark Highlands	8.4.5	68
Central Appalachians	8.4.2	43
Atlantic Coastal Pine Barrens	8.5.4	13
Cross Timbers	9.4.5	1

State	n
GA	67606
AL	55946
SC	48412
MS	42789
TX	39537
NC	33388
AR	30074
VA	26110
LA	22152
FL	6056
TN	4777
OK	3170
MD	2067
DE	661
KY	253
OH	140
WV	126
IL	44
NJ	13
PA	1

Height Growth Prediction

We fit the equation:

$$\Delta ht = ht_{max}\beta_1\beta_2 cr^{\beta_3} e^{(-\beta_1 ht - \beta_4 ccfl - \beta_5 elev + \beta_6 * TD^{0.5} + \beta_7 EMT)}(1 - e^{-\beta_1 ht})^{(\beta_2 - 1)} \tag{1}$$

to the all 383322 growth observations, estimating β_0 through β_7. Where:

- ht = total height (feet)
- cr = crown ratio (fraction of total height)

- $ccfl$ = crown competition factor in larger trees ($feet^2/acre$)
- $elev$ = elevation above sea level (feet)
- TD = temperature difference between minimum and maximum ($^\circ C$)
- EMT = extreme minimum temperature ($^\circ C$)

ht_{max} was estimated by finding the maximum height observed in the data set for loblolly pine. In our data that value is: 162 feet.

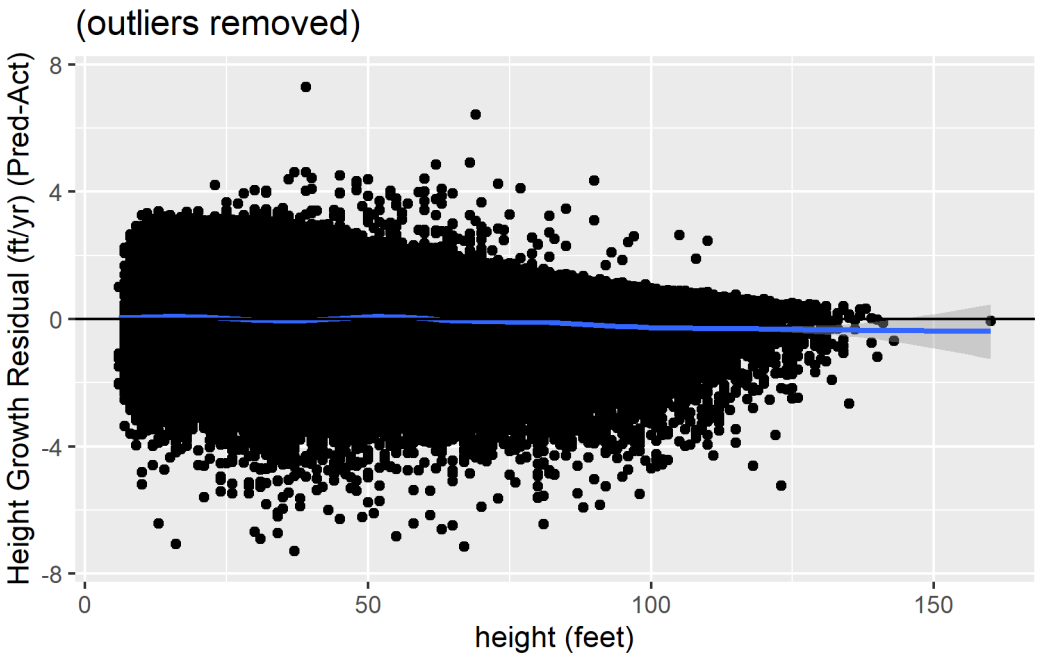
The parameter estimates and fit statistics are:

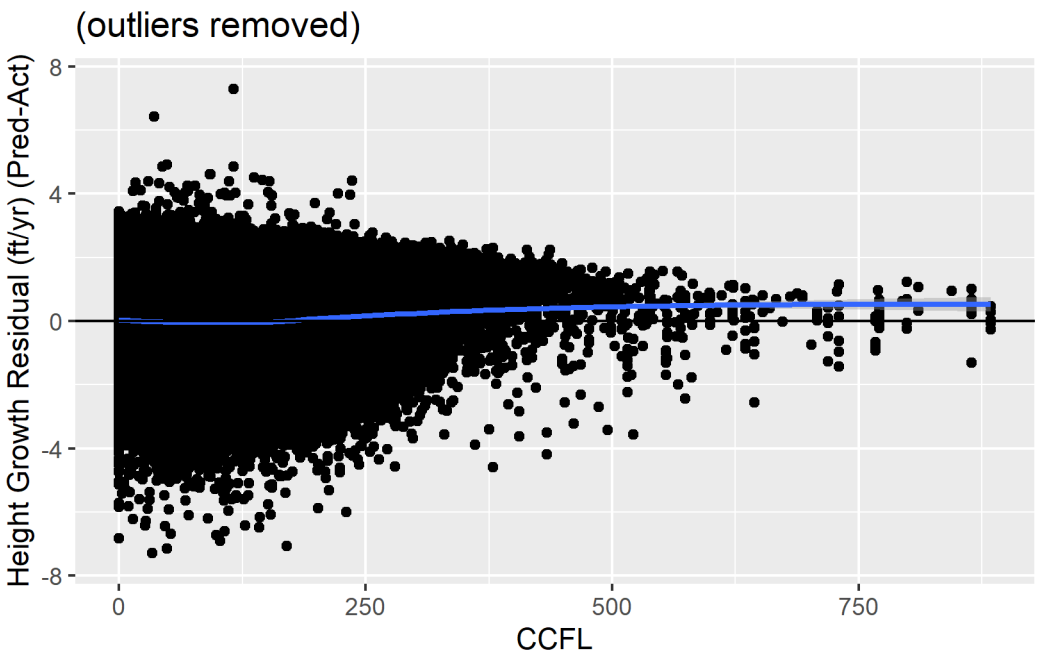
	Coef.	Std. error	t-stat.	p
b1	0.0214633	0.0001167	183.89958	0
b2	1.6584777	0.0099317	166.98788	0
b3	0.2451033	0.0030316	80.85054	0
b4	0.0010686	0.0000138	77.62035	0
b5	0.0001024	0.0000033	30.90919	0
b6	0.2330058	0.0019987	116.57629	0
b7	0.0076808	0.0004244	18.09985	0

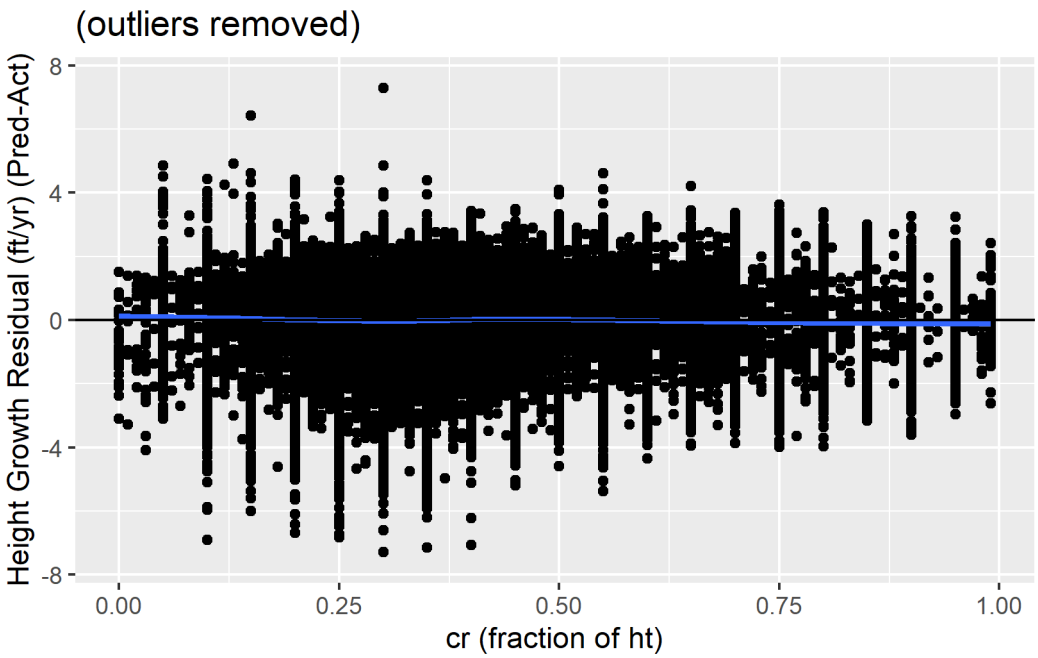
```
Residual Standard Error: 5.98845143166412 on 383315 degrees of freedom. AIC =  2459994.3
```

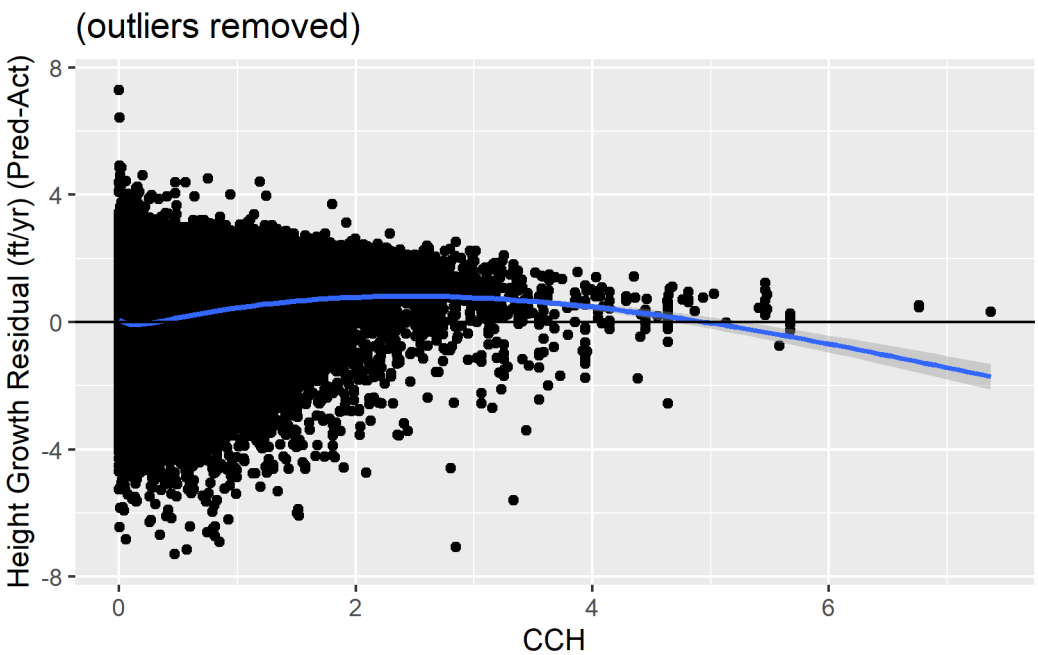
Residual Analysis for Equation 1

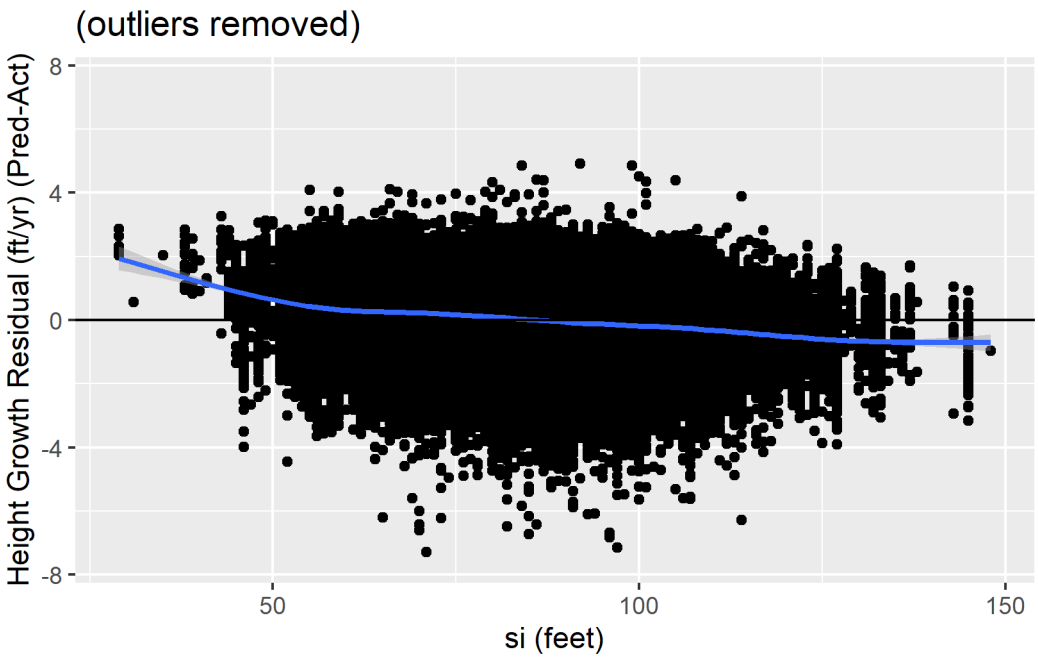
Mensurational Variables

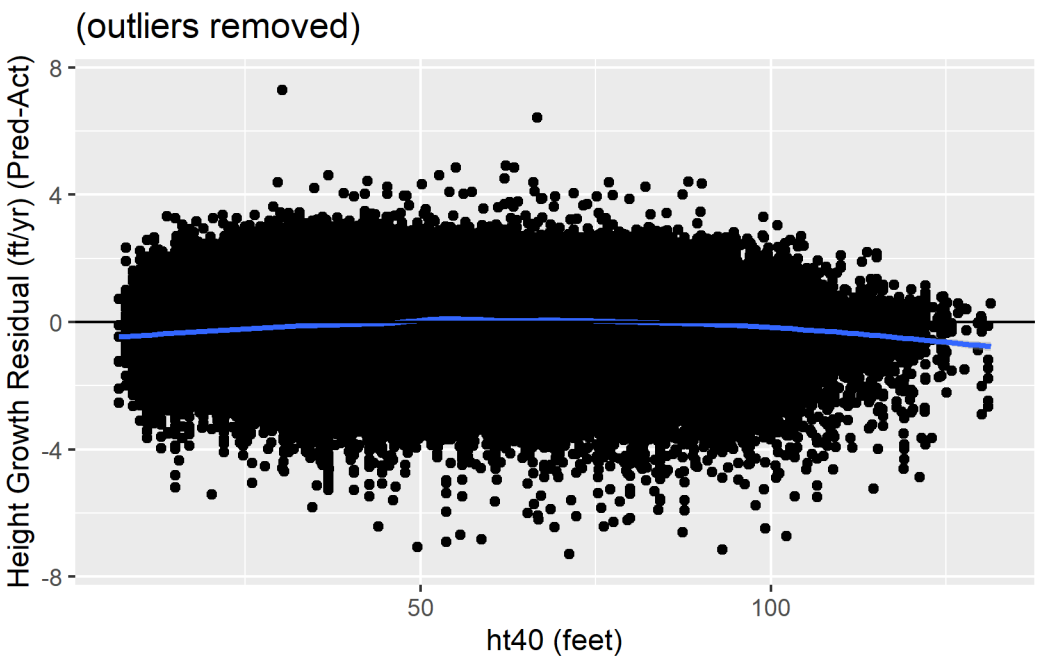




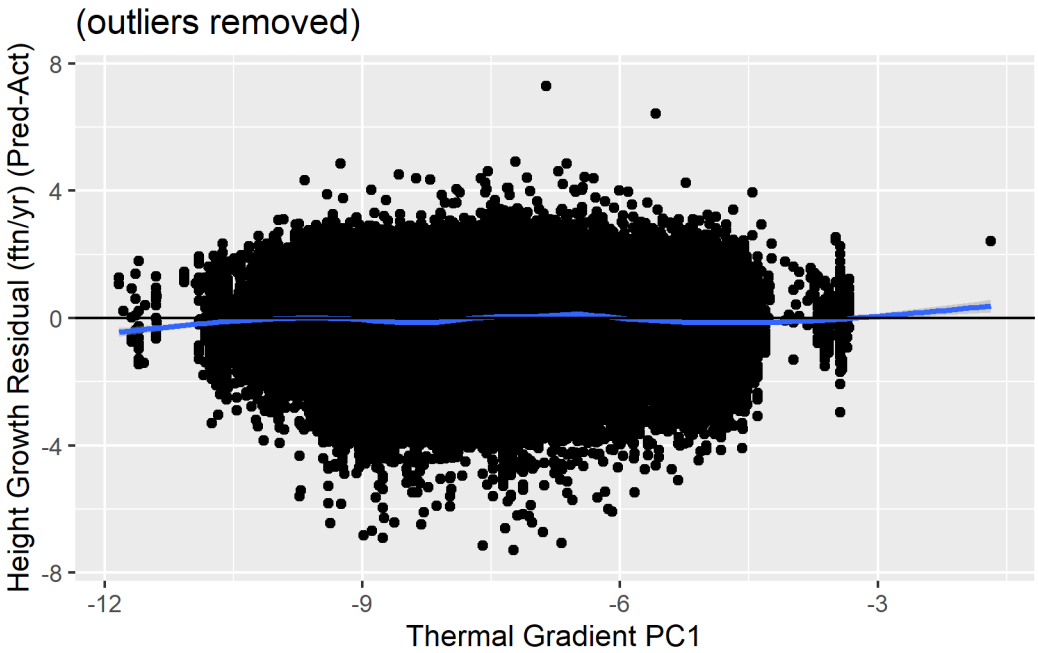

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
0.00
0.25
0.50
0.75
1.00
cr (fraction of ht)

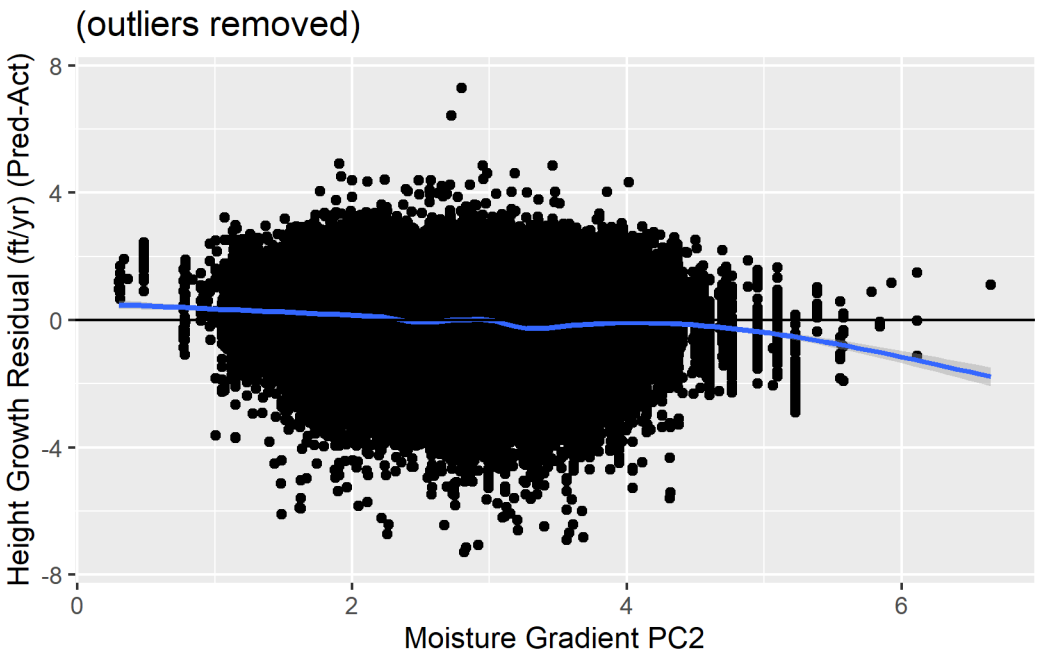

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
0
2
4
6
CCH

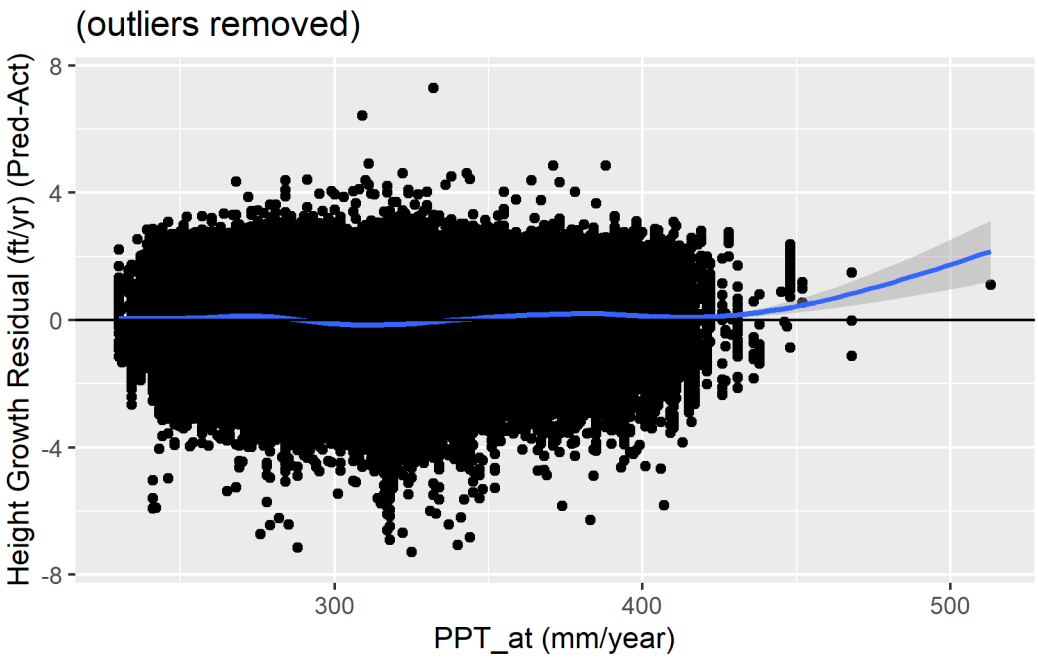

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
50
100
150
si (feet)

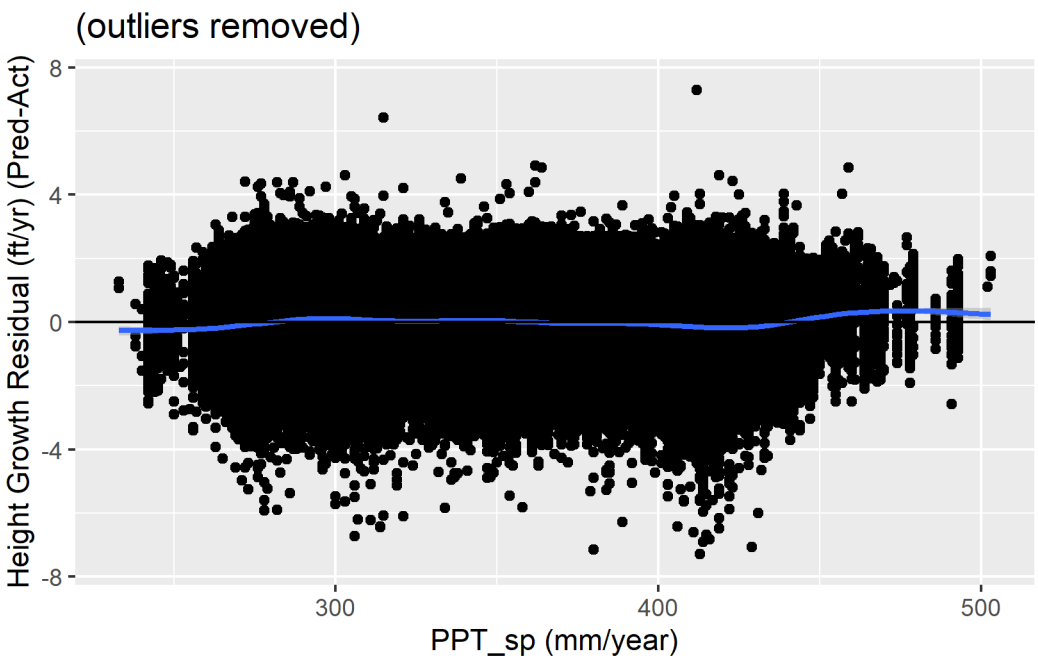

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
50
100
ht40 (feet)

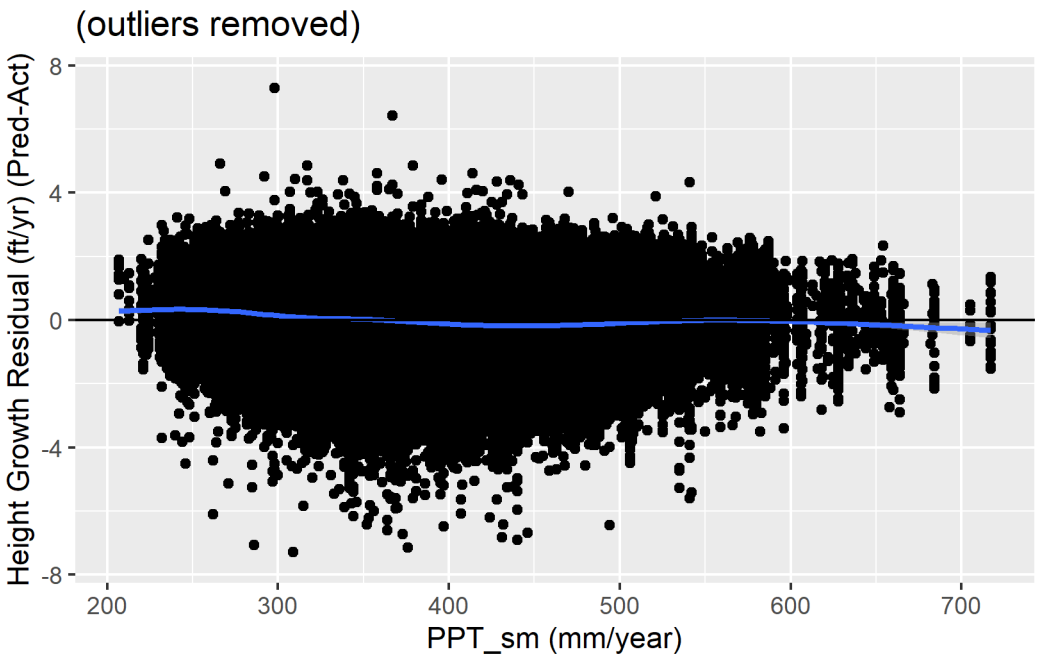
Environmental and Climate Variables

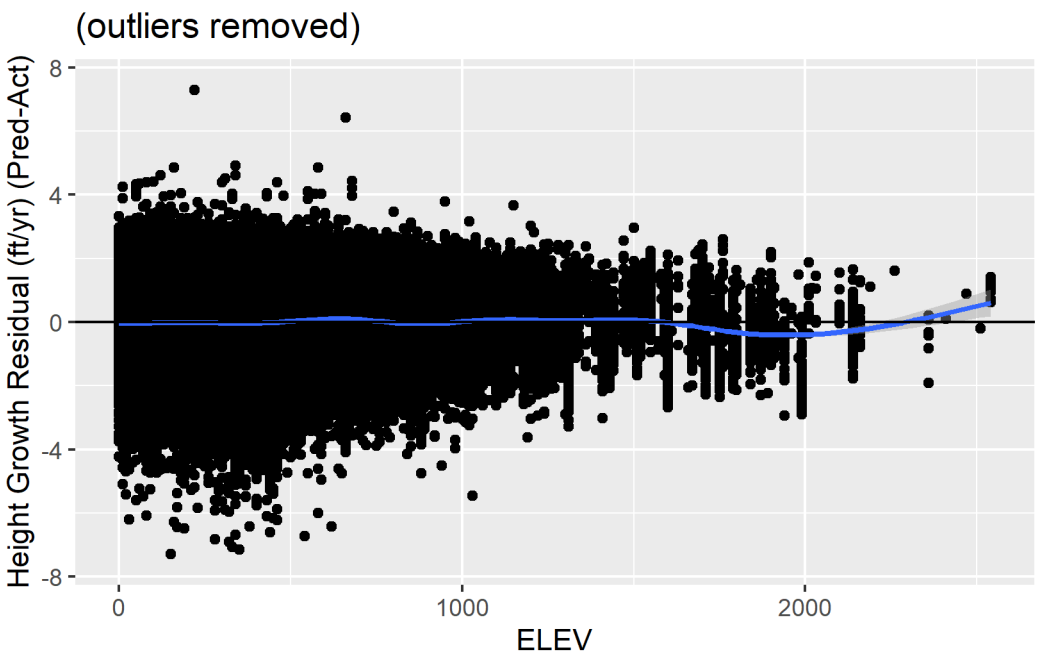


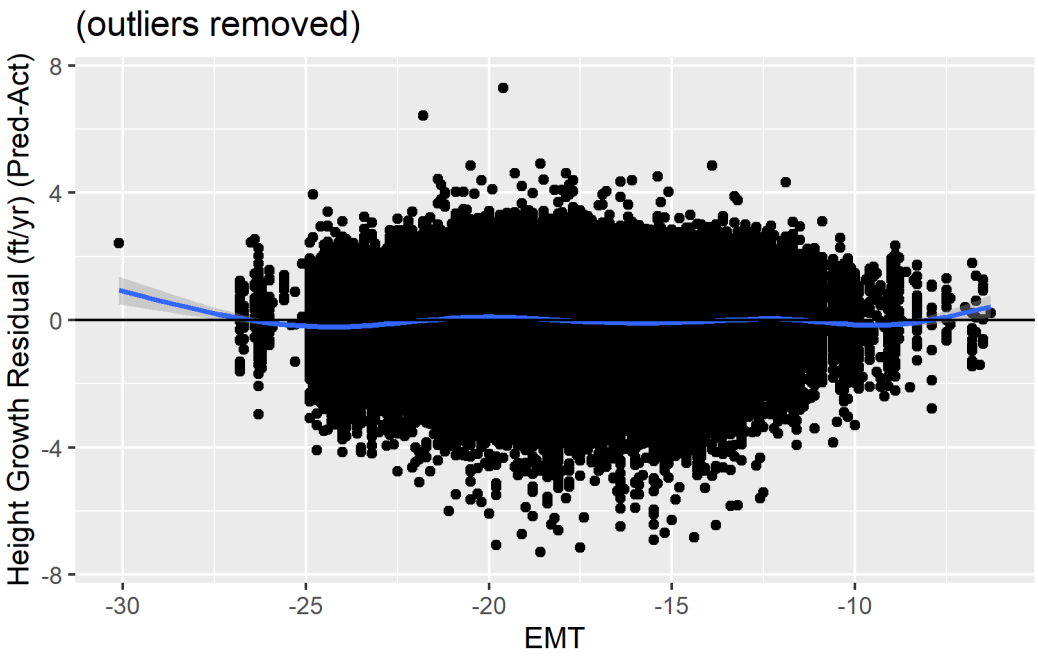


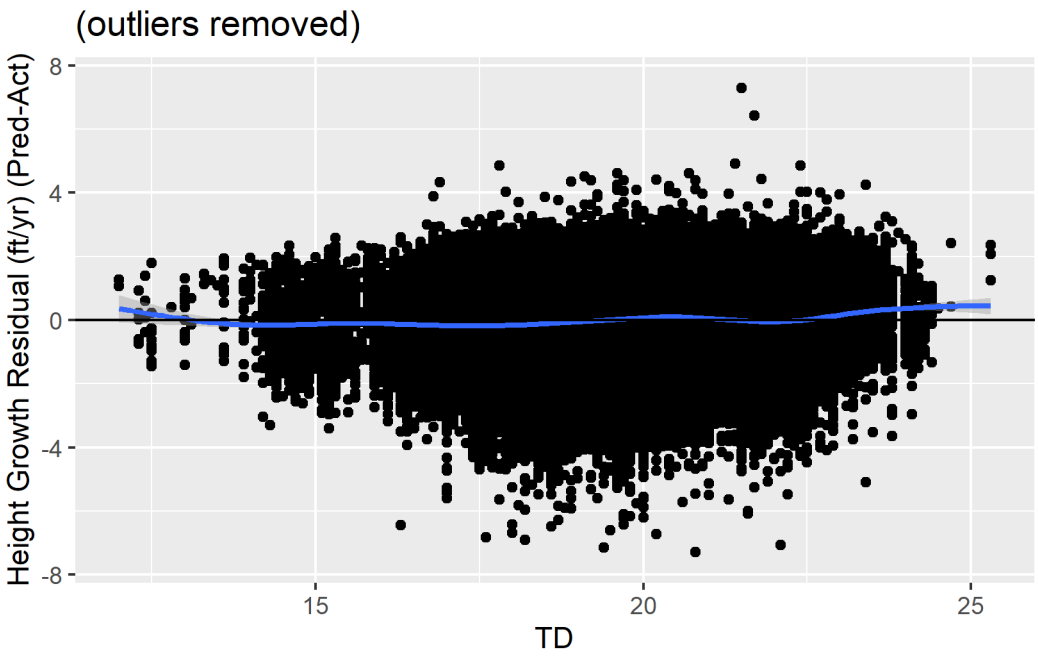

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
300
400
500
PPT_at (mm/year)

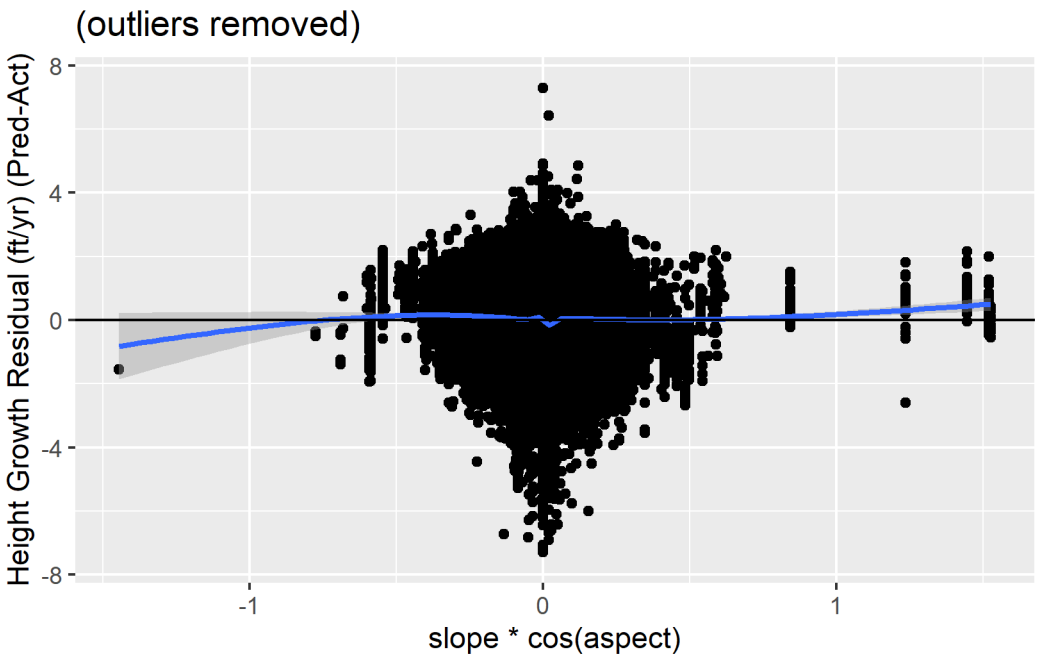

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
300
400
500
PPT_sp (mm/year)

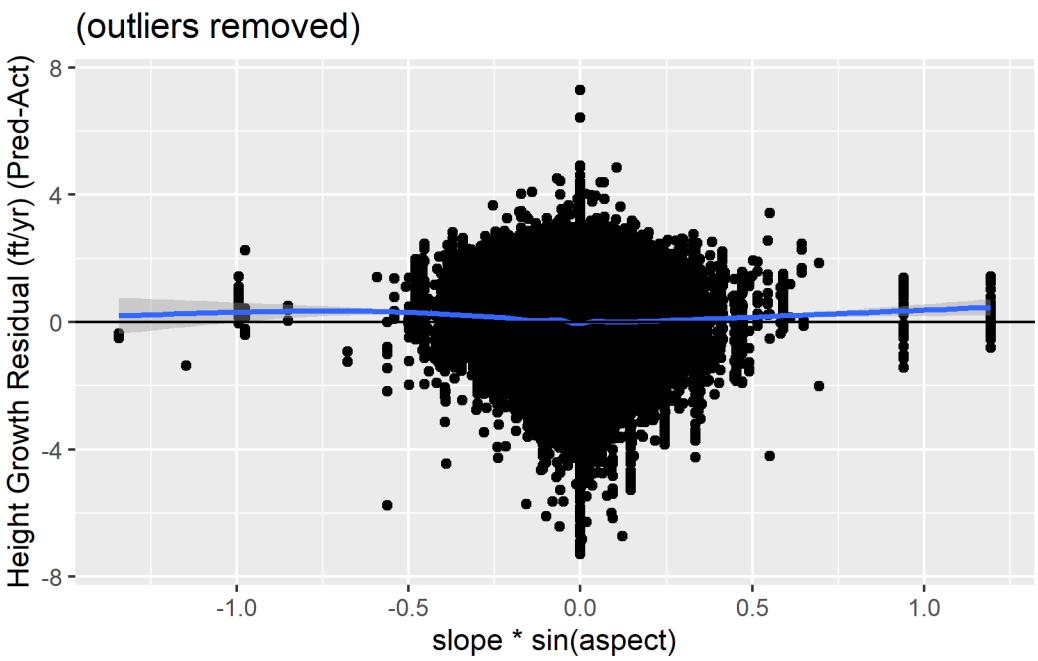

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
200
300
400
500
600
700
PPT_sm (mm/year)


(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
0
1000
2000
ELEV


(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
-30
-25
-20
-15
-10
EMT


(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
15
20
25
TD

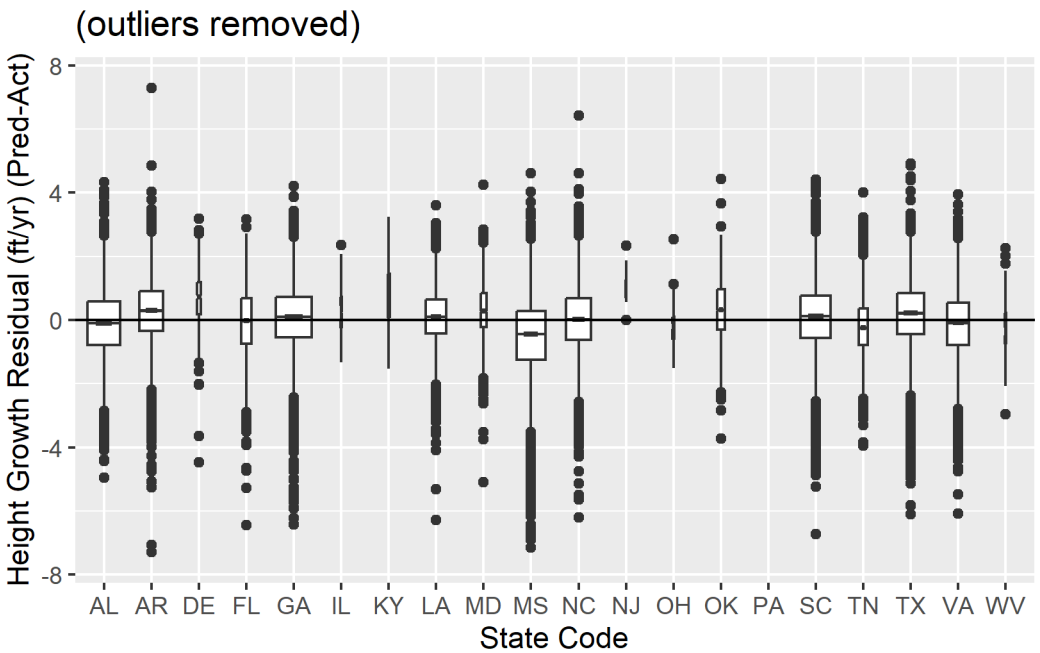


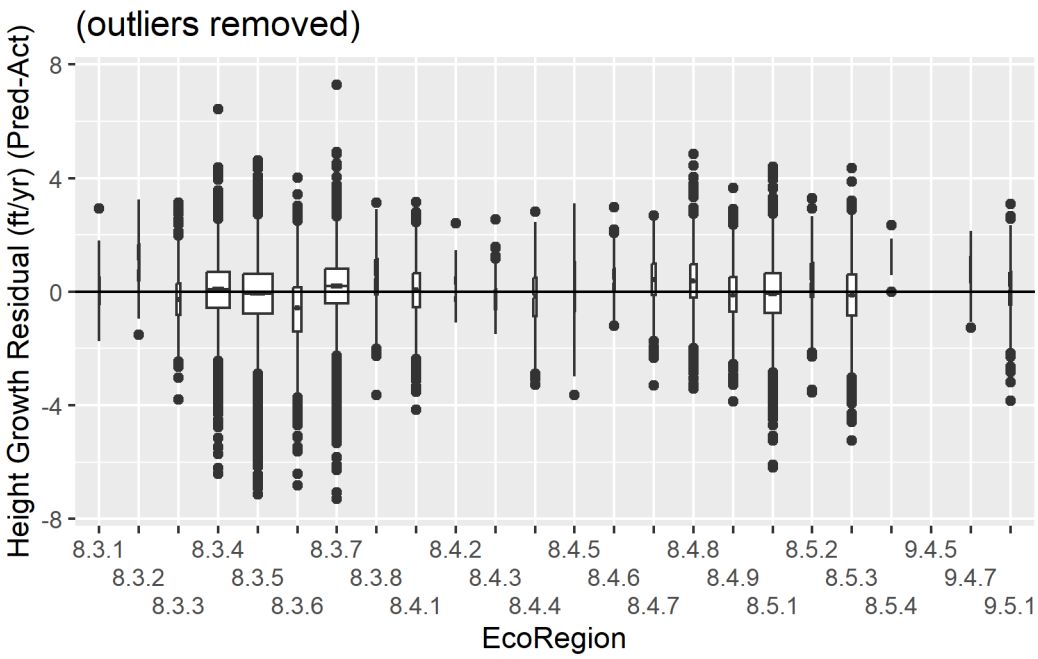


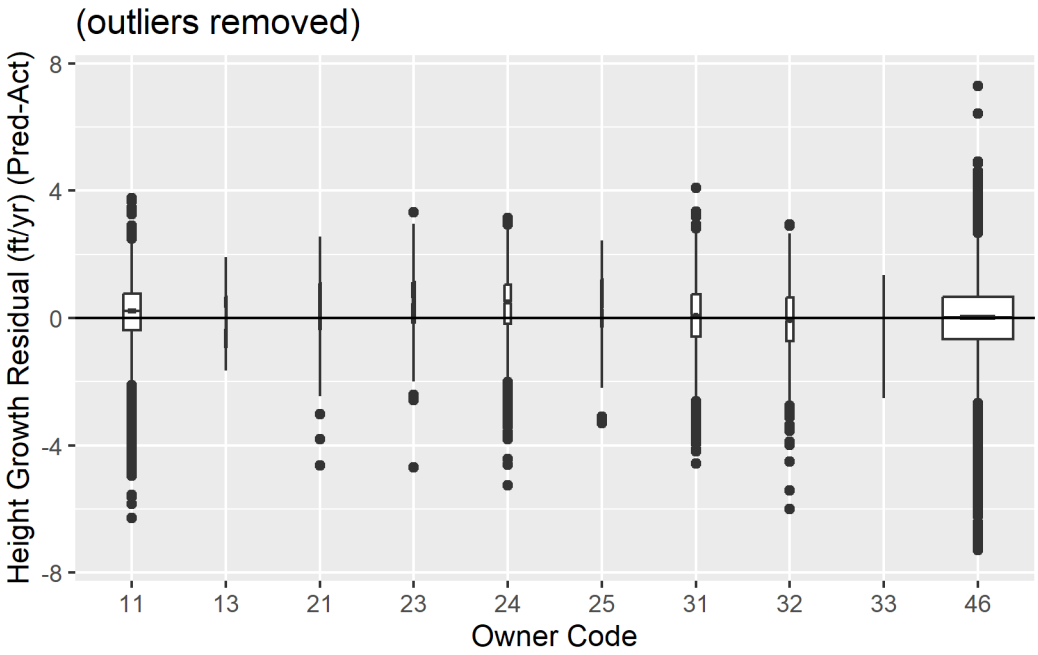

```
     hg_res          AHM         bFFP          CMD          CMI         DD_0
1.000000000  0.098416805  0.019329612  0.093620737 -0.082403309  0.005741310
      DD_18       DD1040         DD18          DD5         eFFP          EMT
```

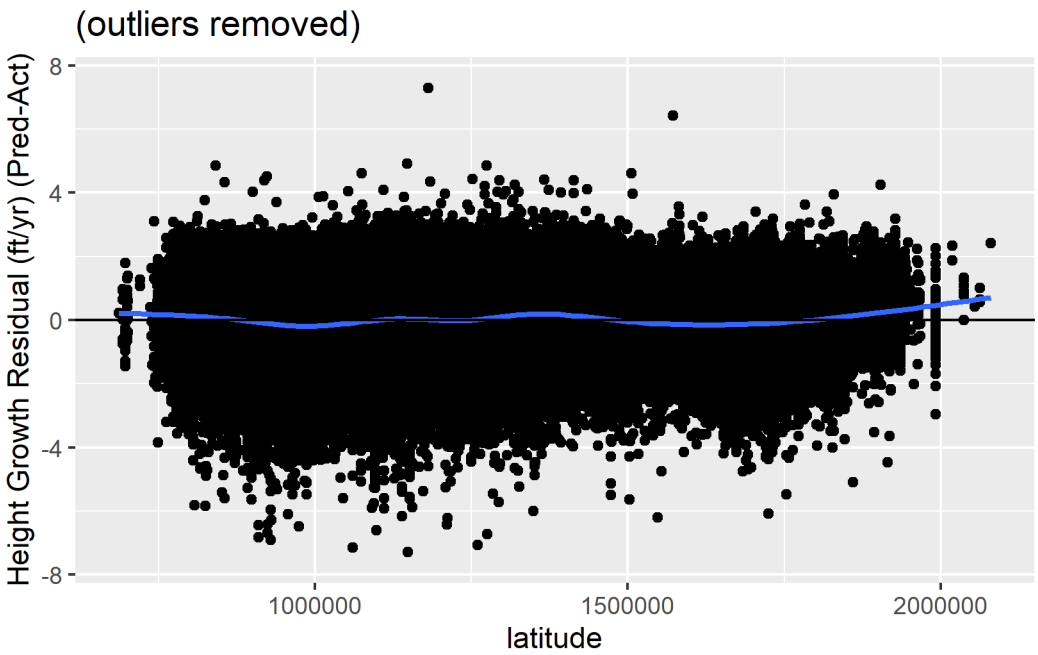
```
 0.015970963 -0.005026538  0.008269968 -0.007610394 -0.013355590 -0.017962445
        Eref           EXT           FFP           MAP           MAT          MCMT
-0.019167723  0.088055279 -0.016956375 -0.102974688 -0.007751467 -0.021461620
         MSP          MWMT          NFFD           PAS        PPT_at        PPT_sm
-0.082071465  0.054318783 -0.018400107 -0.016331728  0.011909937 -0.094548424
      PPT_sp        PPT_wt            RH           SHM       Tave_at       Tave_sm
-0.059220824 -0.074455984  0.025752570  0.103390871 -0.007614722  0.034180468
     Tave_sp       Tave_wt            TD         lyr.1         lyr.2
-0.012055513 -0.023451504  0.060452414  0.023711560 -0.107021327
```

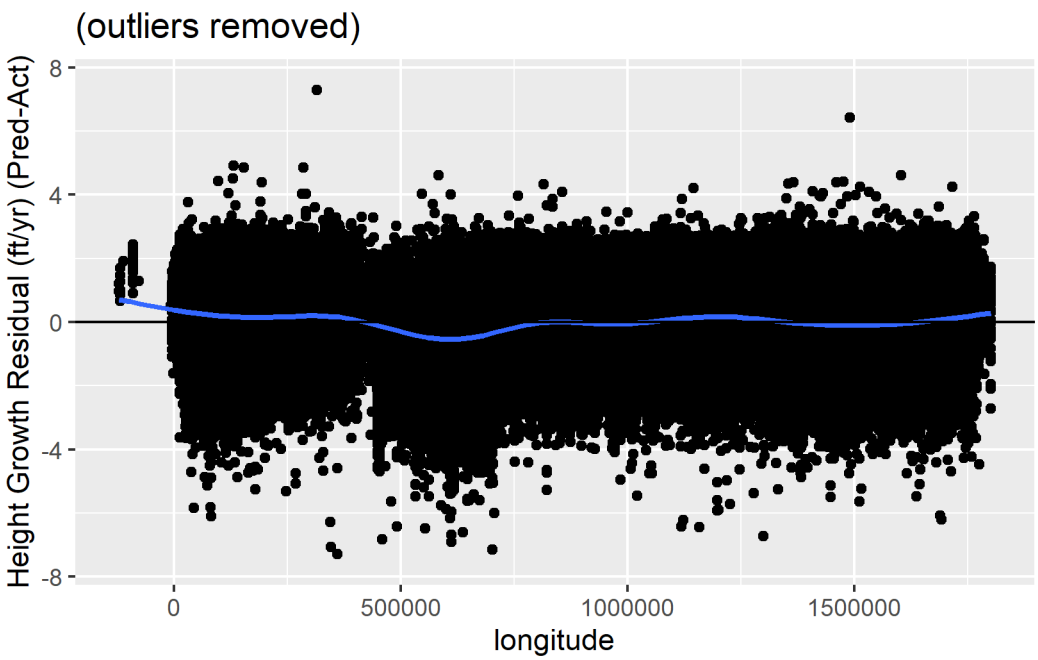
Location Variables

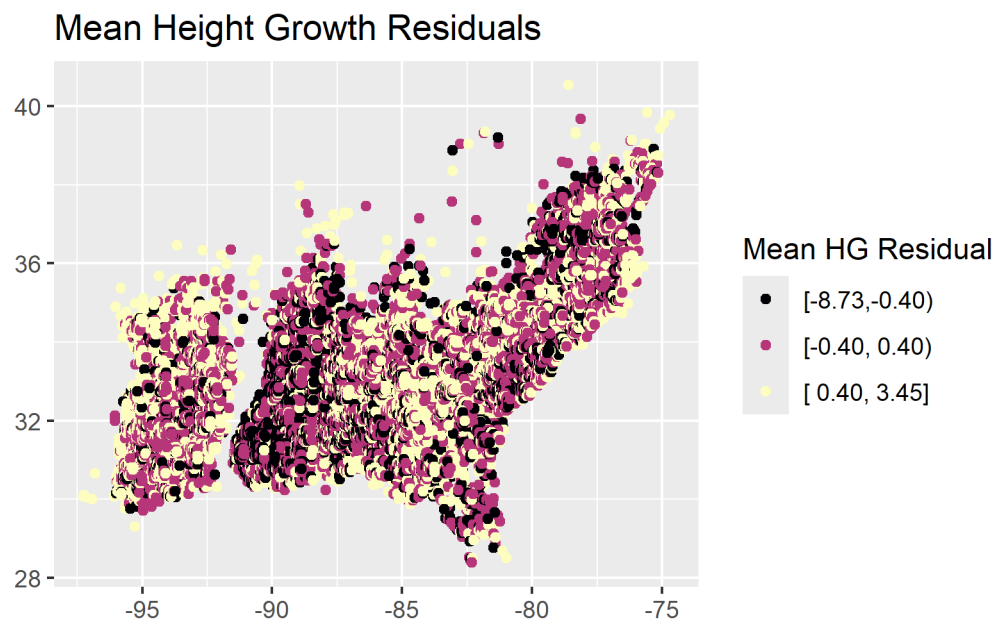



(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
8.3.1
8.3.2
8.3.3
8.3.4
8.3.5
8.3.6
8.3.7
8.3.8
8.4.1
8.4.2
8.4.3
8.4.4
8.4.5
8.4.6
8.4.7
8.4.8
8.4.9
8.5.1
8.5.2
8.5.3
8.5.4
9.4.5
9.4.7
9.5.1
EcoRegion

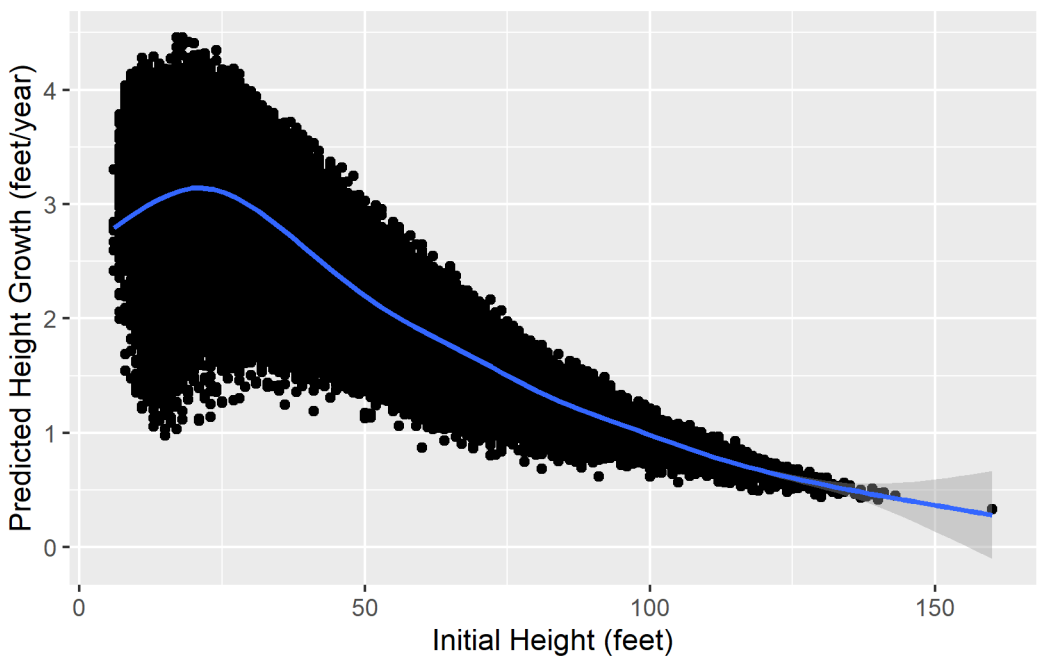

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
11
13
21
23
24
25
31
32
33
46
Owner Code

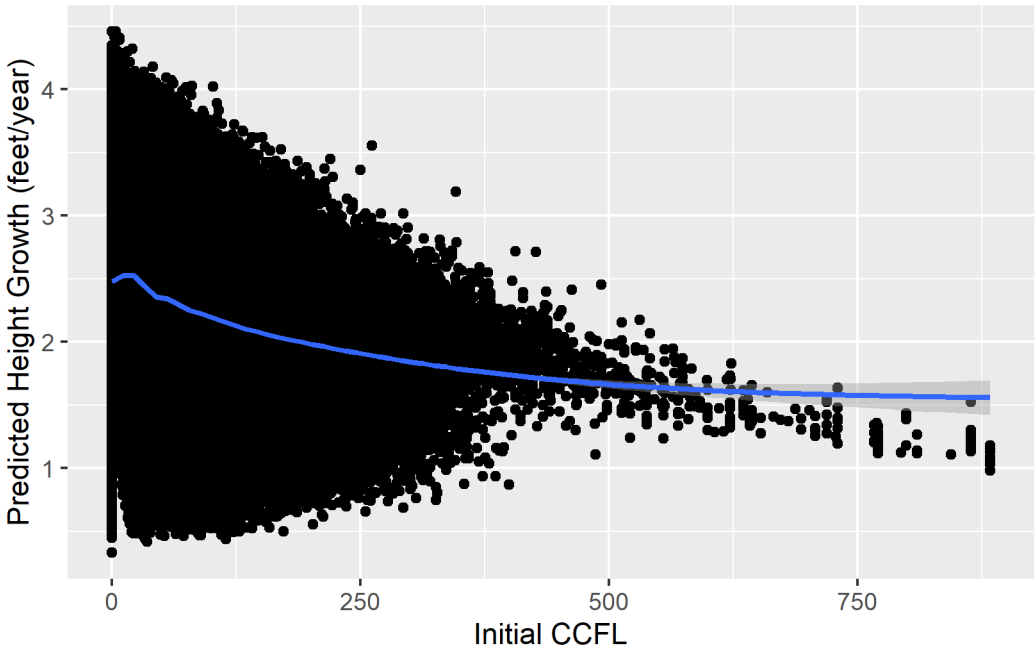

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
1000000
1500000
2000000
latitude

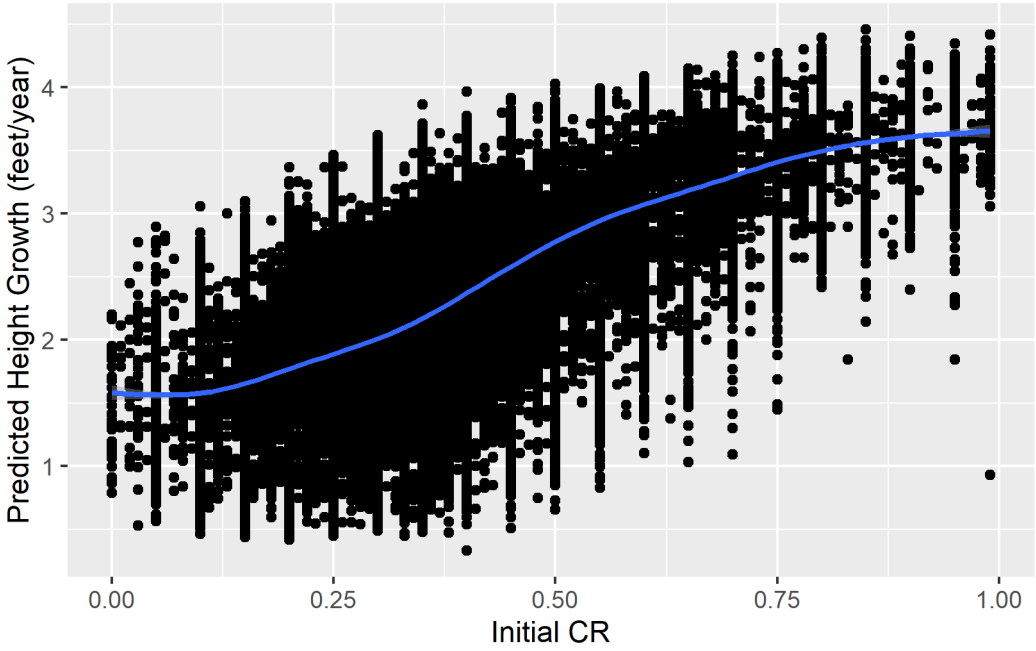

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
8
4
0
-4
-8
0
500000
1000000
1500000
longitude

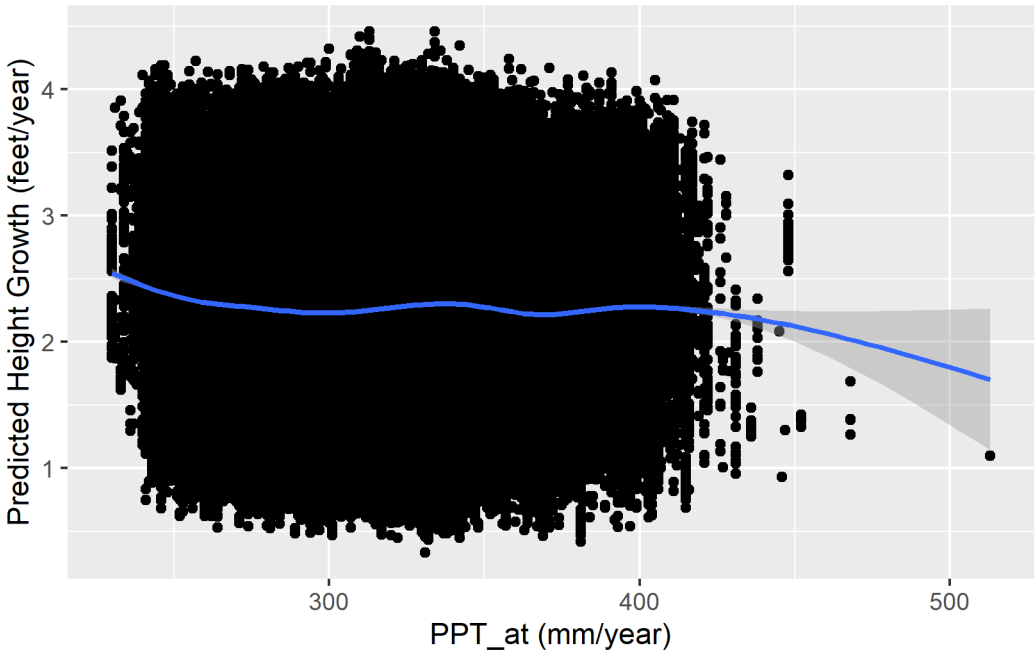


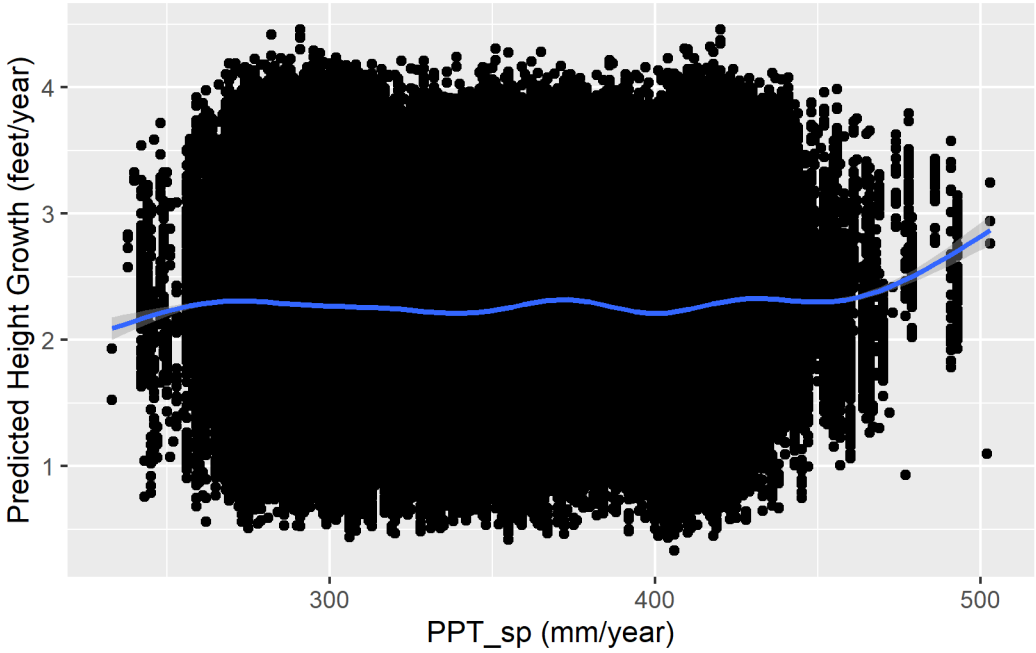
Prediction Performance

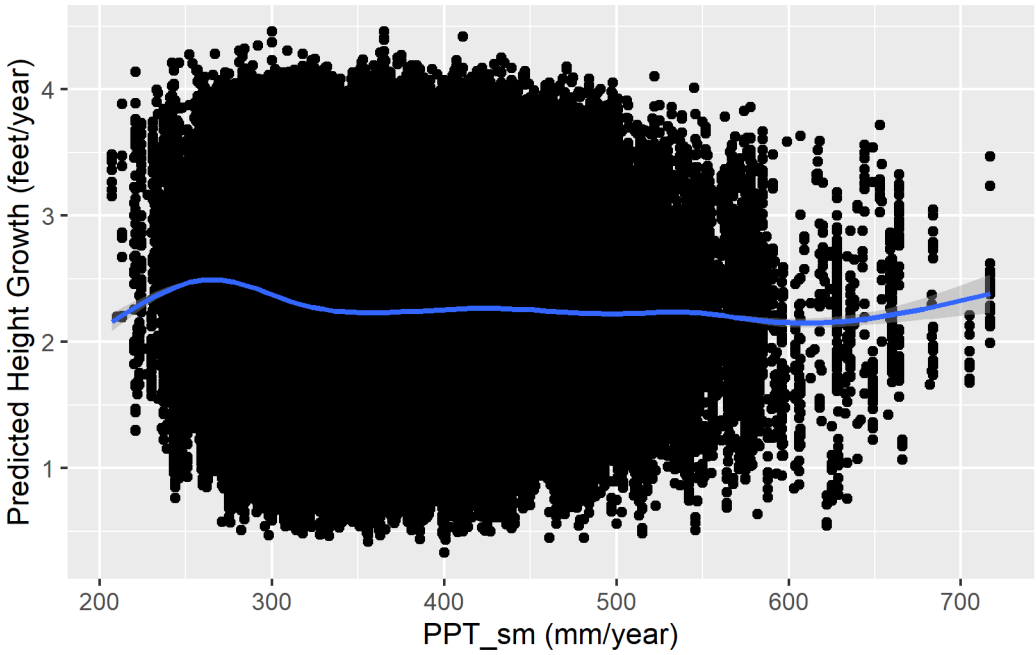


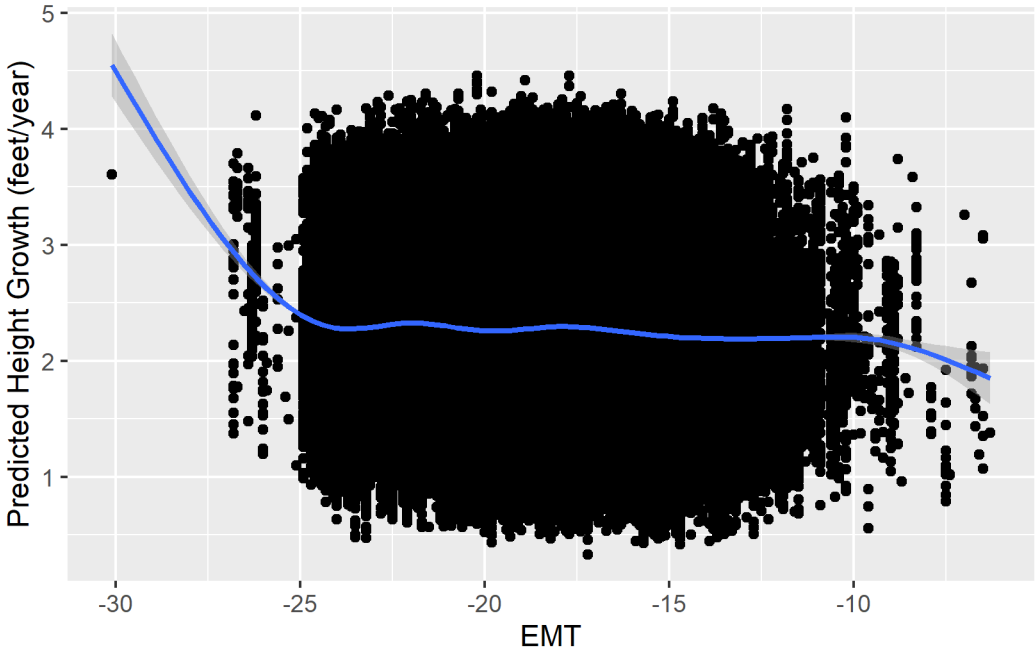

Predicted Height Growth (feet/year)
4
3
2
1
0
250
500
750
Initial CCFL


Predicted Height Growth (feet/year)
4
3
2
1
0.00
0.25
0.50
0.75
1.00
Initial CR


Predicted Height Growth (feet/year)
4
3
2
1
300
400
500
PPT_at (mm/year)


Predicted Height Growth (feet/year)
4
3
2
1
300
400
500
PPT_sp (mm/year)


Predicted Height Growth (feet/year)
4
3
2
1
200
300
400
500
600
700
PPT_sm (mm/year)


Predicted Height Growth (feet/year)
5
4
3
2
1
-30
-25
-20
-15
-10
EMT